Automating Data Import and Relationship Mapping Using Import Sets & Dot Walking

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Abstract

This project focuses on automating the process of importing employee data into ServiceNow and mapping relationships between employees, departments, and managers using Import Sets, Transform Maps, and Dot Walking. The system imports employee records from external CSV files and automatically maps relationships so that related information is displayed when incidents are assigned to users.

Introduction

Organizations often receive employee data from external systems. Manually entering employee records into ServiceNow is time-consuming and error-prone. This project automates the data import process using ServiceNow Import Sets and Transform Maps. Additionally, Dot Walking is used to retrieve related data such as department and manager details directly in the incident form.

Problem Statement

Organizations want to regularly import a list of new employees with details such as name, email, department, manager, and phone number from an external system. When an incident is assigned to an employee, the incident form should automatically display the department and manager details.

Objectives

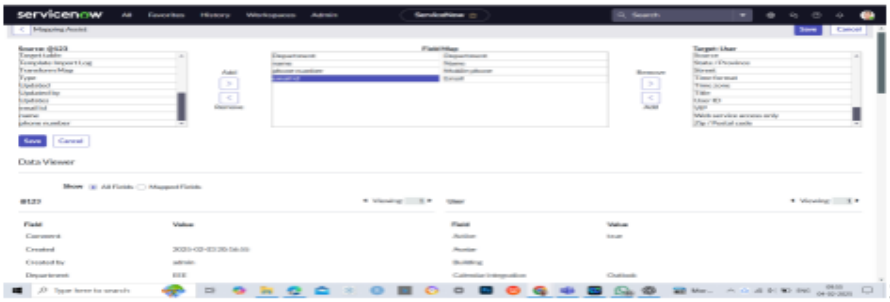
- Import employee records from external CSV files.
- Automatically map employee data to ServiceNow tables.
- Establish relationships between employees, departments, and managers.
- Display department and manager information automatically in incident forms.

System Implementation

The implementation involves creating an employee dataset, importing it into ServiceNow using Import Sets, mapping fields using Transform Maps, and using Dot Walking to retrieve related information.

Implementation Screenshot 1

Screenshot from ServiceNow implementation showing the process step.



3. Enter some records

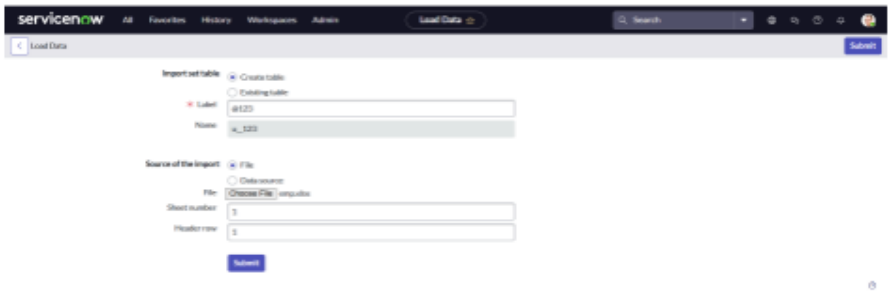
4. Save the sheet as: employee data
5. Download the file as .xlsx , .csv (or) .xml file

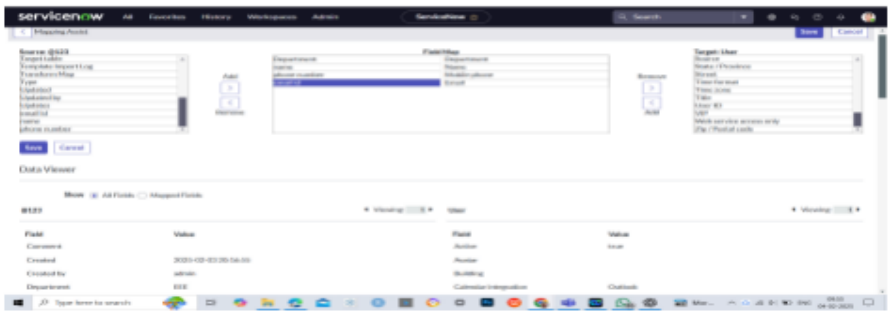
Activity 2: Import data

1. Open service now.
2. Click on All >> search for import sets
3. Select load data under system import sets
4. select create table
5. Give the the import set table name as: import employee
6. Choose the file which we created in the XL sheet.

Implementation Screenshot 2

Screenshot from ServiceNow implementation showing the process step.





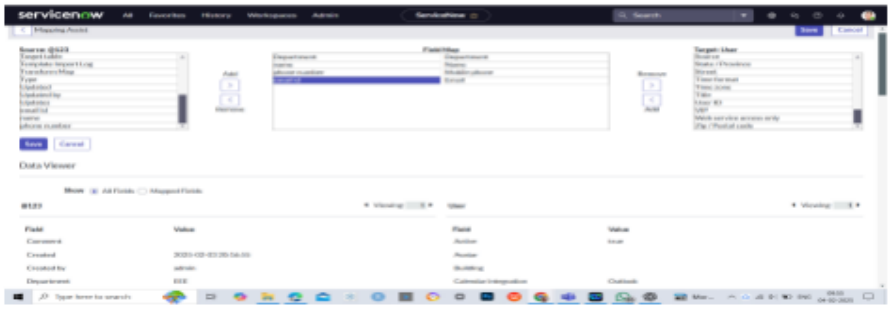
7.The data is imported,click on personalized list
8.Add fields to the form.

Implementation Screenshot 3

Screenshot from ServiceNow implementation showing the process step.

Activity 3: Create transform map

1. Open service now.
2. Click on All >> search for import set
3. Select create transform map under import sets
4. Click on new
5. Fill the following details to create a new transform map
6. Give name as: Test
7. Select source table: @123
8. Select target table: user[sys_user]
9. Save the form.

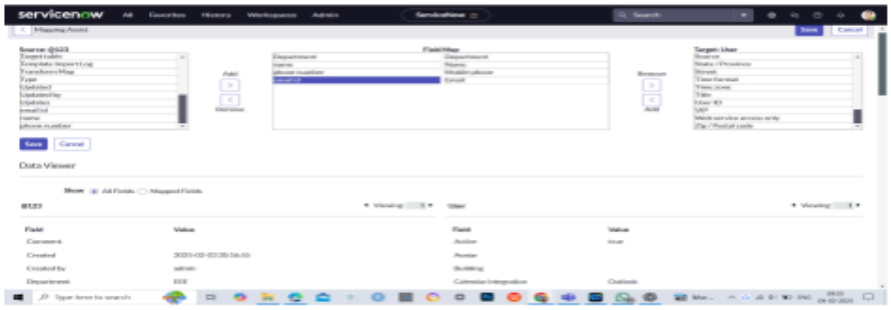


Implementation Screenshot 4

Screenshot from ServiceNow implementation showing the process step.

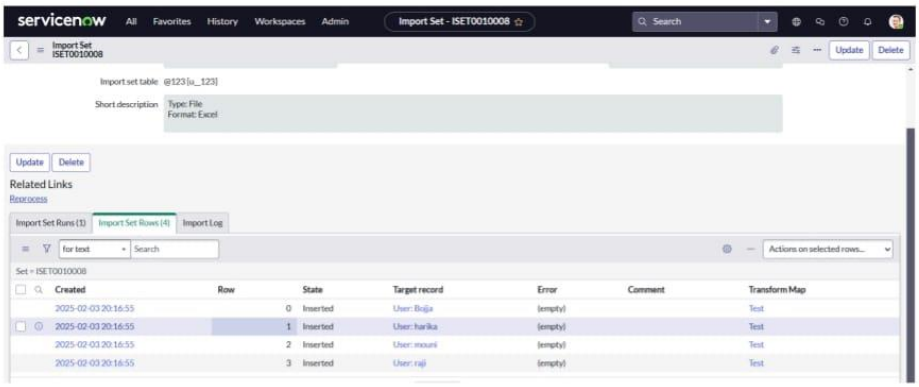
Activity 4: Match the fields using mapping assist

1. Scrolled down the created transform map form
2. Click on auto matching fields
3. Then in the mapping fields some fields are matched



Implementation Screenshot 5

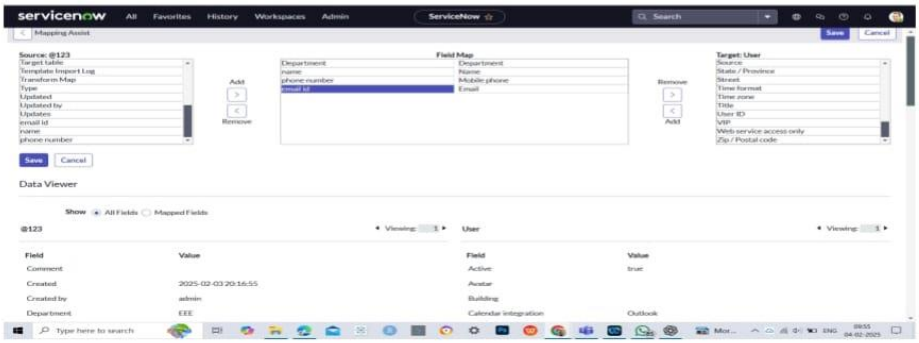
Screenshot from ServiceNow implementation showing the process step.



4. Using mapping assist map the manager field.
5. Select the fields which you are mapping in the staging table and target table
6. Select manager in source table and add
7. Select manager field in target table
8. Click on the save button.

Activity 5: Transform the Data

1. Click on transform in the related links.
2. Click on transform.
3. Data's transform state is completed.



Testing:

The system was tested using multiple employee datasets. Test cases ensured that employee records were correctly imported, mapped to the appropriate fields, and displayed correctly when incidents were assigned.

Test Case	Input	Expected Result	Status
Import CSV	Employee data file	Records loaded into import set	Pass
Transform Map	Imported records	Mapped to user table	Pass
Incident Assignment	Assign user	Department auto display	Pass
Dot Walking	Assign employee	Manager email auto display	Pass

Conclusion:

The project successfully demonstrates how ServiceNow Import Sets, Transform Maps, and Dot Walking can automate data import and relationship mapping. This approach reduces manual work, improves accuracy, and enhances incident management processes.